MapShift: Controlled Post-Intervention Evaluation for Embodied Planning

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Abstract

Embodied agents are often evaluated in the same environment they explored or trained in, making it difficult to tell whether a learned world model supports planning after the world changes. We introduce MapShift, an executable evaluation protocol for controlled post-intervention competence. An agent first explores a base environment without task reward; the environment is then modified by a controlled metric, topology, dynamics, or semantic intervention; and the agent is evaluated on post-intervention planning, inference, and adaptation tasks. The contribution is measurement infrastructure: matched base/intervened pairs, family-wise estimands, automated validity checks, protocol-comparison tooling, and reproducible artifact generation. In the reference study, benchmark health gates pass with zero fatal leakage, zero task rejections, perfect oracle solvability, no intervention-validator failures, and no severity-monotonicity failures. Empirically, MapShift reveals non-redundant family stress profiles, strong sensitivity to exploration assumptions, and a mechanism diagnostic in which controlled post-intervention evaluation changes same-environment conclusions while separating stale memory, local heuristics, and explicit belief update.

1 Introduction

World models are useful only insofar as their learned structure remains actionable when the world does not stay fixed. In embodied evaluation, however, agents are commonly trained, explored, and evaluated under tightly coupled environment assumptions. Even when environments are procedurally generated or randomized, the evaluation often asks whether a system performs well under a sampled distribution, not whether the system can use reward-free exploration to reason about a specific environment after a controlled change. This distinction matters for embodied planning, which combines partial observability and sequential decision-making with structured spatial state [Kaelbling et al., 1998, Sutton and Barto, 2018]. Failures may arise from brittle geometry, missing connectivity, incorrect transition dynamics, or shifted cue meanings.

We study this question as an evaluation problem rather than as a new modeling proposal. The central object is controlled post-intervention evaluation: sample a base environment e, let an agent explore e without task reward for a fixed budget T_{exp}, apply an intervention $I_f(\sigma)$ to obtain a changed environment e', and evaluate the agent on tasks sampled from $Q(e, e')$. The family f identifies what changed, and the severity σ controls how strongly it changed. We use the term "counterfactual" only in this operational matched-pair sense: the generator seed and motif are held fixed while a declared environment factor is changed. The protocol is therefore closer to controlled out-of-distribution evaluation than to unit-level causal identification from observational data. Under this protocol, a method can fail despite strong same-environment performance if its internal representation is not useful after the intervention. Figure 1 summarizes the evaluation flow.

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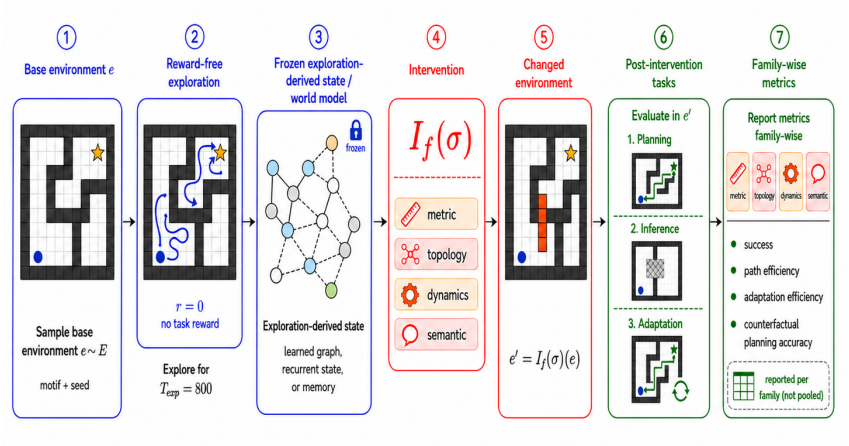

Figure 1: Controlled post-intervention evaluation protocol. An agent first explores a base environment without task reward, producing a fixed exploration-derived memory or world model. A controlled intervention then changes the environment along one family and severity level. The agent is evaluated only after the intervention on planning, inference, and adaptation tasks, and results are reported by intervention family to expose intervention-specific failure modes.

MapShift is a controlled benchmark implementation of this protocol. It is designed as executable measurement infrastructure: it provides matched intervention pairs, family-wise estimands, explicit severity ladders, motif-level splits, reproducible task generation, benchmark health reports, protocol comparisons, and paper-facing artifact builders. MapShift is intended for researchers comparing post-intervention robustness of embodied planners, evaluating new world-model architectures against intervention-family stress profiles, or auditing whether evaluation conclusions depend on protocol assumptions. The substrate is deliberately minimal so that intervention isolation, task solvability, leakage, and weak-baseline saturation can be verified. A similar intervention layer could be built on top of other navigation substrates, including MiniGrid; the contribution here is the substrate-agnostic evaluation methodology and the reference artifact that makes those checks executable.

The paper is organized around three evaluation-science questions rather than a single leaderboard claim. First, what exactly is being measured when an agent is evaluated after an isolated environment change? Second, do different intervention families expose distinguishable failure profiles? Third, when do protocol choices such as reward-free exploration, same-environment execution, and family-wise reporting change conclusions? The reference study gives calibrated answers for one baseline roster: family profiles and exploration assumptions matter, and a deterministic mechanism diagnostic shows that same-environment evaluation can change conclusions about stale maps versus explicit belief update.

Our contributions are:

1. We formalize a controlled post-intervention evaluation protocol for measuring embodied competence after reward-free exploration.

2. We release MapShift, a benchmark with eight structural motifs, four intervention families, four severity levels, three task classes, and a canonical exploration budget of $T_{\text{exp}} = 800$.

3. We provide benchmark health checks covering split leakage, coverage, rejection statistics, oracle solvability, weak-baseline saturation, severity monotonicity, and intervention isolation.

4. We define a reproducible study artifact that compares oracle, same-environment, diagnostic, heuristic, classical belief-update, recurrent, memory, relational, and structured-dynamics baselines across model seeds and protocol variants.

2 Related work

Reinforcement-learning and embodied benchmarks. Benchmark environments such as the Arcade Learning Environment, OpenAI Gym, DeepMind Control Suite, DeepMind Lab, Procgen, and MiniGrid have been central to reproducible reinforcement-learning and agent research [Bellemare et al., 2013, Brockman et al., 2016, Tassa et al., 2018, Beattie et al., 2016, Cobbe et al., 2020, Chevalier-Boisvert et al., 2023]. Crafter and MineDojo extend this line toward open-ended, multi-skill embodied agents [Hafner, 2022, Fan et al., 2022]. Embodied AI simulators and datasets such as Habitat, AI2-THOR, ProcTHOR, VirtualHome, and ALFRED add richer spatial, visual, and task structure [Savva et al., 2019, Kolve et al., 2017, Deitke et al., 2022, Puig et al., 2018, Shridhar et al., 2020]. MapShift is complementary to these efforts: rather than emphasizing simulator breadth, open-endedness, or realistic task semantics, it isolates a specific evaluative question: whether environment knowledge acquired during reward-free exploration remains useful after a controlled intervention. The benchmark should therefore be read as a diagnostic protocol and reference implementation, not as a claim that the substrate itself is uniquely realistic.

World models and planning under partial observability. Model-based reinforcement learning and world-model approaches learn latent or explicit predictive structure for planning and control [Ha and Schmidhuber, 2018, Hafner et al., 2020, Schrittwieser et al., 2020]. Reward-free and reward-agnostic exploration methods separate information gathering from downstream reward optimization [Jin et al., 2020, Sekar et al., 2020, Mendonca et al., 2021]. Classical POMDP formulations make partial observability explicit, while modern embodied benchmarks often test the resulting agents through aggregate task success [Kaelbling et al., 1998]. MapShift focuses on a different diagnostic axis: not only whether a model predicts or plans in its training environment, but whether the representation supports update, inference, and planning after geometry, topology, dynamics, or semantics change following exploration.

Causal and counterfactual reinforcement learning. Causal reinforcement-learning work studies confounding, interventions, and counterfactual reasoning in sequential decision processes [Lu et al., 2018, Buesing et al., 2019, Rezende et al., 2020]. MapShift does not attempt to identify causal effects from observational trajectories, nor does it assume that the agent has learned a structural causal model of the world. Instead, the benchmark generator supplies controlled interventions and matched base/intervened environments. This distinction is important: the evaluation borrows the practical idea of holding background factors fixed while changing an explicit variable, but its claims are about robustness and diagnostic evaluation under known interventions.

Distribution shift and evaluation documentation. Distribution-shift benchmarks such as WILDS emphasize that aggregate in-distribution performance can obscure robustness failures under changed conditions [Koh et al., 2021]. Ranking analyses of challenge benchmarks show that leaderboard conclusions can be sensitive to aggregation choices, test-case resampling, and the metric used to compare rankings [Maier-Hein et al., 2018, Perlitz et al., 2024]. Recent automated benchmark work similarly uses rank-correlation agreement between generated and human-labeled evaluations as evidence for evaluation reliability [Chen et al., 2024]. Dataset and model documentation work argues that evaluation artifacts should state their intended use, scope, limitations, and provenance [Gebru et al., 2021, Mitchell et al., 2019]. MapShift differs from MiniGrid-style procedural sampling by constructing matched base/intervened pairs for a fixed explored environment; from WILDS-style supervised shift benchmarks by evaluating embodied state acquired through reward-free exploration before intervention; and from causal RL by using known generator interventions for diagnostic robustness rather than estimating causal effects under confounding.

3 Controlled post-intervention protocol

Protocol. Let $\mathcal{E}$ denote a distribution over embodied environments, $\mathcal{F}$ the set of intervention families, and Σ_f the severity ladder for family f. A base environment $e \sim \mathcal{E}$ is sampled and exposed to the agent for reward-free exploration. During exploration, the system M receives observations and actions but no downstream task reward. We write the resulting internal state as

$$z_M = \mathrm{Explore}(M, e, T_{\mathrm{exp}}), \tag{1}$$

where z_M may be a recurrent hidden state, an explicit memory, a learned graph, or an empty state in the no-exploration ablation. After exploration, an intervention operator $I_f(\sigma)$ is applied, where $f \in \mathcal{F} = \{\text{metric}, \text{topology}, \text{dynamics}, \text{semantic}\}$ and $\sigma \in \Sigma_f$ is a discrete severity level. The resulting environment is

$$e' = I_f(\sigma)(e). \tag{2}$$

A task $q \sim Q_c(e, e')$ is then sampled from task class $c \in \mathcal{C}$, where $\mathcal{C}$ contains planning, inference, and adaptation. The evaluated system is scored using z_M from exploration in e while solving or inferring in e':

$$S(M; e, f, \sigma, c, q) = \text{Perf}(M, z_M, e', q)\,. \tag{3}$$

The population target of the benchmark is therefore

$$\text{CEP}(M) = \mathbb{E}_{e\sim\mathcal{E},\, f\sim\mathcal{F},\, \sigma\sim\Sigma_f,\, c\sim\mathcal{C},\, q\sim Q_c(e,e')}\left[S(M; e, f, \sigma, c, q)\right]. \tag{4}$$

In the paper, we do not treat this expectation as a single scalar leaderboard. The main estimand is family-wise:

$$\text{CEP}_f(M) = \mathbb{E}_{e,\, \sigma\in\Sigma_f,\, c,\, q}\left[S(M; e, f, \sigma, c, q)\right], \tag{5}$$

because the benchmark is designed to separate metric, topology, dynamics, and semantic failure modes.

Operational intervention semantics. The generator can be viewed as a deterministic map $G(u, x)$, where u contains motif identity, sample seed, and layout randomness, and $x = (x_{\text{metric}}, x_{\text{topology}}, x_{\text{dynamics}}, x_{\text{semantic}})$ contains the configurable environment factors. A family intervention constructs

$$e' = G(u, x_{-f}, \phi_f(x_f, \sigma))\,, \tag{6}$$

where x_{-f} denotes the non-target factors and ϕ_f is the family-specific transformation. This is the sense in which the benchmark creates matched post-intervention pairs: the background motif and seed are held fixed, while one declared factor is changed and validators check that non-target factors remain invariant. We do not use the evaluation to infer causal effects from observational data, and we do not claim latent-unit counterfactual identification. The protocol is a controlled test-time shift with explicit intervention provenance.

Empirical estimator. The executable study estimates $\text{CEP}_f(M)$ with a finite set of motifs, seeds, severities, task samples, and model seeds. Let $\mathcal{D}_f$ be the set of valid episode records for family f and baseline M, after task rejection. The reported family-wise score is

$$\widehat{\text{CEP}}_f(M) = \frac{1}{|\mathcal{D}_f|} \sum_{r\in\mathcal{D}_f} s_r, \tag{7}$$

where s_r is the task-normalized primary score for episode record r.

Protocol comparison. Protocol sensitivity is measured by replacing pieces of the evaluation operator while holding the generated study grid fixed. Same-environment evaluation replaces e' with e at task execution time. No-exploration evaluation replaces z_M with an empty memory. Short- and long-horizon evaluations multiply task horizons by fixed factors. Rank changes are computed from the induced orderings

$$\pi_p(f) = \text{rank}_M\left(\widehat{\text{CEP}}_{f,p}(M)\right), \tag{8}$$

where p indexes the protocol variant. We summarize protocol sensitivity using Kendall τ, rank reversals, best-method changes, and family-wise rank changes.

Why post-intervention evaluation. The protocol separates two questions that are often conflated. Same-environment evaluation asks whether a system can exploit a representation in the environment it observed. Post-intervention evaluation asks whether that representation remains useful under a specified controlled change. This is close to the diagnostic question behind world-model evaluation [Ha and Schmidhuber, 2018, Hafner et al., 2020, Schrittwieser et al., 2020]: what structure did the model learn, and how does that structure fail?

4 Methodology

Executable study pipeline. The benchmark study is generated from versioned configuration files rather than from manually curated episode lists. For each motif and sample seed, the generator produces a base environment e with a deterministic identifier and manifest. Each baseline then receives the same reward-free exploration budget in e. Learned baselines convert their exploration trace into a local graph-learning problem and train or load a checkpoint keyed by baseline, environment identifier, seed, and hyperparameter hash. For each intervention family f and severity σ, the corresponding operator constructs $e' = I_f(\sigma)(e)$ and records both the changed environment and the invariants that should have been preserved. The task sampler then draws planning, inference, and adaptation tasks from $Q(e, e')$. Impossible tasks and trivial tasks are rejected before model evaluation and counted in the health report.

Intervention construction. Table 1 summarizes the four intervention families. Each family changes one intended axis while preserving other axes when possible. Severity level 0 is the identity operator and is included as a sanity check; levels 1–3 increase the configured severity parameter. Severity is an ordinal within-family ladder, not a cross-family physical unit: severity 2 in topology is not claimed to be equally difficult to severity 2 in dynamics. The chosen ladders span small, medium, and large changes for each family and are interpreted through health checks for oracle solvability, weak-baseline non-saturation, intervention isolation, and within-family severity-response monotonicity.

Table 1: Intervention methodology. Each family has a scalar severity parameter, preservation constraints, and expected failure modes used for benchmark health checks and interpretation.

Family	Severity parameter	Values	Preserved quantities
Metric	action-gain scale	1.0, 1.1, 1.25, 1.5	topology, semantic identities
Topology	blocker fraction	0.0, 0.08, 0.18, 0.32	semantic identities, action response
Dynamics	friction multiplier	1.0, 0.9, 0.75, 0.6	geometry, semantic identities
Semantic	remap fraction	0.0, 0.1, 0.25, 0.5	geometry, action response

Figure 2 shows the matched-pair construction used by these interventions.

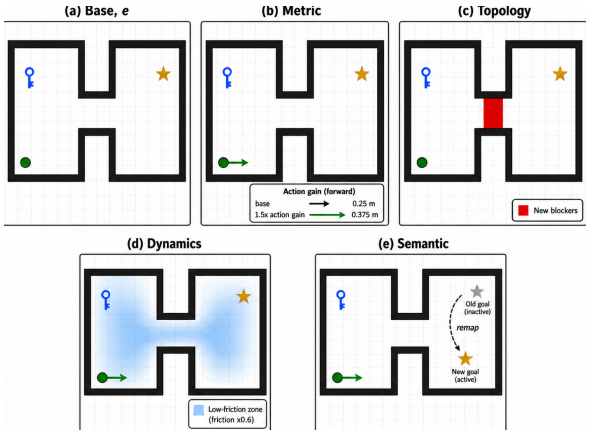

Figure 2: Matched intervention pairs in MapShift. A single base environment (a, two-room connector motif) is shown alongside the same environment after high-severity (level 3) interventions from each of the four families: metric (b, action-gain rescaling), topology (c, new corridor blockers), dynamics (d, low-friction zones), and semantic (e, goal-token remap). Background motif, seed, key location, and agent start are held fixed across panels; only the declared environment factor changes. Validators check that non-target factors remain invariant after each intervention. Severity-3 levels are shown for visibility; the full study uses severities 0, 1, 2, and 3 per family.

Environment and observation model. The reference substrate uses 96×96 occupancy maps at 0.25m resolution. The action interface is semi-continuous navigation with a 0.25m forward primitive

and 15° turn primitive. Observations are egocentric and local: the agent observes a 3.5m radius, 100° field of view, and semantic channels when visible. Exploration uses a single episode, uniform navigable starts, and no privileged global state for non-oracle baselines.

Exploration budget. The exploration budget is held fixed at $T_{\text{exp}} = 800$ steps for all motifs and non-oracle baselines. We do not normalize the budget by motif size or graph diameter because exploration insufficiency is itself part of the evaluated condition: a representation that works only after near-complete coverage should be distinguishable from one that remains useful under partial coverage. The benchmark health report therefore includes visited-cell ratios, visited-node ratios, path-length distributions, and weak-baseline saturation checks, so that under-exploration in larger motifs is visible rather than hidden by the protocol.

Intervention validation. After each intervention is applied, validators check whether the intervention changed the intended factor and preserved the declared invariants. For example, metric shifts should preserve topology and semantic identities; topology shifts should preserve cue identities and action-response parameters; dynamics shifts should preserve geometry; and semantic shifts should preserve navigable geometry and reachability. These validators are used as health checks rather than as model-facing information. Failed intervention validations are reported before model results and block a valid release artifact.

Task sampling. The three task classes are sampled under shared policies that hold fixed the start-state distribution, goal distribution, visibility distribution, and episode-horizon distribution across comparable conditions. Planning tasks include shortest-path navigation, rerouting after blockage, changed-dynamics navigation, and changed-cue navigation. Inference tasks include topology-change detection, masked-region prediction, and counterfactual reachability queries. Adaptation tasks include limited post-shift interaction followed by replanning and few-shot replanning after action-gain changes. Planning horizons are 64, 96, and 128 steps; inference horizons are 16 and 32; adaptation budgets and horizons are 16, 32, and 64.

Task rejection and solvability. The sampler rejects tasks that cannot be solved by the oracle under the intended environment, tasks whose answer is unchanged in a way that makes the post-intervention query vacuous, and tasks whose shortest path or horizon makes the episode trivial. Rejections are not silently discarded: the artifact records rejection counts by reason and by benchmark cell. The health report therefore distinguishes a genuinely difficult benchmark cell from a cell that was under-covered because valid tasks could not be sampled.

Baseline evaluation. All non-oracle baselines share the same $T_{\text{exp}} = 800$ exploration budget. The oracle post-intervention planner receives privileged access to the changed environment and serves as a solvability/reference check rather than a deployable method. The same-environment reference is a protocol control: it evaluates what would be concluded if tasks were executed in the explored environment rather than the intervened environment, and should not be interpreted as a deployable upper-bound method. A stale-map planner is included as a diagnostic sensitivity floor: it stores the pre-intervention map and intentionally does not revise it after the intervention, so it should perform well only when stale exploration-derived state remains valid. A classical belief-update planner stores an occupancy map and token bindings from exploration, receives the same task specification and limited post-shift interaction budget as other non-oracle methods, and updates its belief from local post-shift mismatch observations; it does not receive the intervention label, privileged mismatch locations, or the global post-intervention map. The learned baselines use fixed hyperparameters and five model seeds for the main study: monolithic recurrent, persistent memory, relational graph, and structured dynamics. Each episode record stores the scientific baseline name, concrete run identifier, model seed, protocol, motif, split, family, severity, task class, task type, and environment/model-seed grouping key.

Weak heuristic floor. The weak heuristic is intended as a meaningful but limited floor. During exploration it stores visited cells, visited node identifiers, visible goal-token bindings, and a shallow summary of local traversable structure. For planning, it does not reconstruct or repair a global map. It selects the remembered goal token if available, then follows an orthogonal visible-path planner on remembered local structure; paths that are too long for the horizon, invalid after the intervention, or directed to stale semantic targets fail. For topology-change inference it always predicts

"no topology change," making it intentionally poor at structural revision. For semantic reachability queries it returns the remembered pre-intervention token binding, exposing stale-memory failures. For masked-region queries it predicts the intervention family from simple task-type rules rather than learned evidence. For adaptation, it receives the same limited post-shift budget as other non-oracle methods and reports a hand-coded recovery curve capped below oracle performance unless its planning proxy succeeds. This makes the heuristic stronger than random guessing but too weak to saturate the benchmark if the tasks require genuine post-intervention reasoning.

Information access. Non-oracle methods receive only their exploration trace, task specification, and internal memory; the global map and intervention label are reserved for explicit reference baselines.

Scoring. Planning episodes record success, observed path length, oracle path length, normalized path efficiency, oracle gap, and post-intervention planning correctness. For a planning record r, path efficiency is

$$\eta_r = \begin{cases} \min(\ell_r^*/\ell_r, 1) & \text{if the agent succeeds and } \ell_r > 0, \\ 0 & \text{otherwise,} \end{cases} \tag{9}$$

where ℓ_r^* is the oracle path length in the evaluation environment and ℓ_r is the observed executed length. Post-intervention planning accuracy is the success rate on planning tasks whose correct answer depends on the intervention, excluding identity shortest-path sanity tasks.

Inference episodes record predicted and expected answers and are scored by task type. For masked-region prediction, the target is the changed family label or masked state category, and accuracy is exact-match:

$$A_{\text{mask}} = \frac{1}{N_{\text{mask}}} \sum_{r=1}^{N_{\text{mask}}} \mathbf{1}[\hat{y}_r = y_r]. \tag{10}$$

For post-intervention reachability queries, the target is the post-intervention node or room associated with a changed semantic token; the score is exact-match reachability/location accuracy,

$$A_{\text{reach}} = \frac{1}{N_{\text{reach}}} \sum_{r=1}^{N_{\text{reach}}} \mathbf{1}[\hat{g}_r = g'_r], \tag{11}$$

where g'_r is the correct target in e'. For topology-change detection, records have binary labels $y_r \in \{0, 1\}$ and model scores c_r. Probabilistic baselines may report calibrated confidence; deterministic baselines use $c_r = 1$ for a predicted change and $c_r = 0$ otherwise. AUROC is computed from average ranks with ties:

$$\text{AUROC} = \frac{\sum_{r:y_r=1} \text{rank}(c_r) - n_+(n_+ + 1)/2}{n_+ n_-}. \tag{12}$$

Adaptation episodes record a recovery curve over the allowed post-shift interaction budget and summarize it with adaptation sample efficiency. The family primary score is computed from task-relevant components and is used for ranking only after family-wise aggregation. Confidence intervals use 1000 bootstrap resamples at 95% confidence, grouped by environment/model-seed unit so that repeated task records from the same environment/model-seed combination are not treated as independent.

5 MapShift benchmark design

MapShift uses a partially observed navigation substrate with occupancy, transition, and semantic layers. The design exposes enough structure to manipulate geometry, connectivity, dynamics, and semantics independently, while remaining cheap enough for many seeds, protocol comparisons, and benchmark health checks. Its purpose is not to compete with richer simulators on realism; it is to make the intervention and validation machinery inspectable. The release contains eight structural motifs: simple loop, two-room connector, branching chain, asymmetric multi-room chain, offset bottleneck, nested bottleneck, deceptive shortcut, and disconnected subregion. Splits are defined at the motif level, reducing the chance that evaluation rewards memorization of a generated map family. The three held-out test motifs support leakage-controlled protocol comparison, but not broad motif-level generalization claims; the primary estimand is family- and protocol-level behavior over the configured study grid.

Table 2: MapShift release values. These values define the main study and artifact contract.

Component	Value	Component	Value
Map size	96×96	Exploration budget	$T_{\text{exp}} = 800$
Motifs	8	Splits	3 train, 2 val, 3 test
Families	metric/topology/dynamics/semantic	Severities	0, 1, 2, 3
Task classes	planning/inference/adaptation	Task samples	3 per class/cell
Learned seeds	5 per learned baseline	Bootstrap	1000 grouped resamples

6 Benchmark health and validity checks

An evaluation benchmark should be evaluated before its models are. The MapShift artifact therefore generates a benchmark health report before writing model results. The report includes motif split counts, family/severity/task coverage, task rejection statistics, oracle solvability, weak-baseline non-saturation, path-length and horizon distributions, severity monotonicity, intervention isolation, and leakage checks.

Leakage is reported in three categories. Fatal leakage corresponds to split or task leakage that invalidates a result and must be zero. Diagnostic warnings identify suspicious patterns requiring inspection. Benign reusable templates are explicitly labeled with rationale, such as shared query forms that do not reveal held-out environments. In the completed run, fatal leakage, leakage warnings, intervention-validator failures, severity-monotonicity failures, and task rejections are all zero, while oracle solvability is 1.000.

7 Experimental program

Baselines. The study suite reports method classes plus protocol and diagnostic references: an oracle post-intervention planner, a same-environment reference condition, a weak heuristic, a classical belief-update planner, a monolithic recurrent world model, a persistent-memory world model, a relational graph world model, a structured-dynamics world model, and a stale-map diagnostic planner used in the mechanism diagnostic. The same-environment condition is included to compare conclusions under a different evaluation assumption; it is not interpreted as a deployable method. The deterministic methods have no trainable parameters. The learned baselines are not pretrained across environments; each is trained self-supervised from the exploration trace of the current base environment, then cached and reused for protocol variants involving the same environment and model seed. Training targets are derived from the exploration-time graph representation: edge existence, normalized geometric cost, normalized traversal cost, and semantic-token location. The shared loss is the sum of edge binary cross-entropy, geometry mean-squared error, traversal mean-squared error, and token binary cross-entropy.

Learned baseline mechanisms. The monolithic recurrent baseline applies a GRU to the sequence of node features and predicts pairwise edges, costs, traversal costs, and token locations from hidden states. The persistent-memory baseline uses learned memory slots: node features attend to a bank of slots, and the slot readout is used for the same prediction heads. The relational graph baseline encodes node features and performs message passing over the explored adjacency matrix before prediction. The structured-dynamics baseline factorizes the model into a geometry encoder and a dynamics encoder; edge and geometry predictions depend on geometry factors, while traversal-cost prediction additionally receives global dynamics features. Full hyperparameters and parameter counts are given in Appendix B.

Model seeds and aggregation. Learned baselines are run with five model seeds. Hyperparameters are specified in versioned configuration files before final evaluation; validation motifs are reserved for pre-final configuration choices, and test motifs are not used for hyperparameter selection. Records include both the scientific baseline name and the concrete baseline run identifier, avoiding the common mistake of collapsing model seeds into a single run. Summaries average over model seeds and bootstrap over environment/model-seed units.

Study grid. The main study covers all eight motifs, all four intervention families, severities 0–3, planning/inference/adaptation tasks, and three task samples per class. It reports the primary CEP table as methods by intervention family, then severity-response curves and protocol sensitivity analyses.

Protocol comparisons. To test whether evaluation assumptions change conclusions, we compare CEP to same-environment evaluation, no-exploration evaluation, short-horizon evaluation, and long-horizon evaluation. The reported statistics include Kendall τ, rank reversals, best-method changes, family-wise rank changes, pooled-versus-familywise ranking disagreement, and leave-one-family-out ranking sensitivity. These statistics are designed to evaluate conclusions, not merely scores.

8 Empirical calibration findings

The reference run should be read as a calibration of the measurement infrastructure, not as a definitive leaderboard. Health checks are strong: task rejections are zero, oracle solvability is 1.000, fatal leakage is zero, intervention-validator failures are zero, and severity-monotonicity failures are zero. The weak heuristic obtains a non-saturating health success estimate of 0.721, distinct from the family-wise primary CEP scores in Table A.5, indicating that tasks are accessible to simple strategies but not solved by the floor baseline.

The empirical findings are substantive but not overstated. MapShift shows that evaluation conclusions depend on which post-intervention assumption is changed: removing reward-free exploration strongly alters rankings, family-wise reporting exposes distinct stress profiles, and same-environment versus CEP pooled rankings are stable for the full learned-probe roster. Intervention-family signatures differ, severity responses are monotone in the health report, and baseline behavior varies across semantic, topology, metric, and dynamics shifts. The learned baselines are lightweight trace-trained probes rather than competitive embodied agents. In the completed run, the hand-coded weak heuristic outperforms learned probes on metric, topology, and dynamics, while the recurrent probe is strongest among non-oracle methods on semantic shifts. The persistent-memory probe collapses to near-zero on three families, suggesting that 16 learned slots are insufficient to retain navigable structure under this single-environment training regime. We interpret these patterns as calibration: MapShift exposes when simple spatial priors dominate naive learned world models and when protocol choices change conclusions. Full family-wise scores and confidence intervals are in Appendix A. Removing reward-free exploration changes all four family-wise method orderings, with Kendall $\tau = 0.524$ and five rank reversals relative to CEP.

A separate deterministic mechanism diagnostic isolates stale memory, local heuristic priors, and explicit belief update. It compares oracle, stale-map, weak-heuristic, and classical belief-update baselines on the same generated intervention grid. Under CEP, belief update scores 0.513/0.689/0.699/0.838 on metric/topology/dynamics/semantic families, compared with stale-map scores of 0.438/0.447/0.771/0.228 and weak-heuristic scores of 0.442/0.542/0.687/0.218. Because the family primary score aggregates planning, inference, and adaptation components, the oracle post-intervention planner is a solvability and planning reference rather than an upper bound for every mixed family score. The pattern is diagnostic: stale maps remain competitive when geometry is preserved, but explicit update recovers much of the gap under topology and semantic shifts. On this deterministic subset, same-environment evaluation and CEP disagree substantially: Kendall $\tau = 0.333$, with two rank reversals, three family-wise rank changes, and a best-method change. This complements the full-roster result, where pooled same-environment rankings remain stable but family-conditioned orderings change.

MapShift therefore audits several conclusions that a standard or underspecified protocol could invite: stale maps can look competitive under same-environment evaluation, learned memory slots can look like a sufficient architectural answer, and no-exploration evaluation can appear close enough. Under CEP, these become more precise diagnoses: updating rather than stale reuse is the target competence, the persistent-memory probe does not retain usable navigable structure in this regime, and exploration assumptions are load-bearing. Appendix Table A.1 gives the compact conclusion audit.

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Appendix

A Additional result tables

Table A.1: Conclusion audit enabled by MapShift. Standard or underspecified protocols invite the left-hand conclusion; MapShift supports the middle diagnosis.

Standard conclusion	MapShift diagnosis	Why it matters
Same-env pooled order is enough.	Full-run pooled order is stable, but the belief-update diagnostic changes best method ($\tau = 0.333$).	Standard evaluation can reward stale state.
Stale maps are competitive.	Same-env ranks stale-map too highly; CEP favors belief update under topology/semantic shifts.	Updating, not reuse, is the target competence.
Learned memory is robust.	Persistent memory is near zero on three families.	Slots alone do not retain usable structure.
No-exploration is close enough.	All four family orders change ($\tau = 0.524$, five reversals).	Exploration assumptions are load-bearing.

Table A.2: Benchmark health summary for the completed full run. Health checks are generated before model-result interpretation and included in the release artifact.

Check	Value
Environments and splits	8 motifs; 3 train, 2 validation, 3 test
Task coverage	1152 tasks; 0 task rejections
Oracle solvability	1.000
Weak-baseline health estimate	0.721 success proxy; not primary CEP score
Fatal leakage / benign warnings	0 / 0
Intervention validator failures	0
Severity monotonicity failures	0

Table A.3: Protocol sensitivity in the full run. Ranking comparisons are computed from the same generated study grid while changing one evaluation assumption.

Comparison	Kendall τ	Rank reversals	Family changes
No exploration vs. reward-free exploration	0.524	5	4
Same-environment vs. CEP	1.000	0	2
Short vs. long horizon	1.000	0	3
Pooled vs. family-wise reporting	1.000	0	0

B Baseline metadata

Training source. Learned baselines are trained separately for each base environment and model seed. They are not pretrained across environments and they do not train on generated maps outside the current exploration instance. The training tensors are built from the exploration-derived graph: node features contain normalized position, local free/blocked/semantic observation statistics, role indicators, and global dynamics features; pair labels supervise edge existence, normalized geometric path cost, and normalized traversal cost; node labels supervise semantic-token location.

Optimization and hyperparameters. All learned baselines use Adam with learning rate 0.01. Pair and node supervision tensors are split into train and validation masks with a validation fraction of 0.2, and early stopping uses patience 3. Hyperparameters are specified in configuration files before the final run; validation motifs are the only intended place for pre-final configuration choices. Test motifs are not used for hyperparameter selection. Checkpoints are keyed by baseline name, environment id, model seed, and hyperparameter hash, then reused across protocol variants.

Compute reporting. The smoke path is designed to run on CPU and completes in approximately 2–5 minutes for the smoke build itself; the one-command artifact audit takes longer because it also

Table A.4: Protocol sensitivity in the deterministic mechanism diagnostic. This focused run compares oracle, stale-map, weak-heuristic, and classical belief-update baselines.

Comparison	Kendall τ	Rank reversals	Family changes	Best changes
No exploration vs. reward-free	0.667	1	1	No
Same-env. vs. CEP	0.333	2	3	Yes
Pooled vs. family-wise	0.667	–	2	Yes

Table A.5: Main family-wise CEP calibration results from the full run. Entries are primary family scores. Learned baselines are lightweight self-supervised probes trained only from each exploration trace; the weak heuristic is a hand-coded floor with strong local spatial priors. The same-environment reference and oracle planner share deterministic shortest-path mechanics in this study and are reported separately to support protocol-comparison interpretation. Full 95% grouped-bootstrap confidence intervals are included in the artifact JSON and rendered Markdown tables.

Method	Metric	Topology	Dynamics	Semantic
Oracle planner	0.480	0.745	0.771	0.807
Same-env. reference	0.480	0.745	0.771	0.807
Weak heuristic	0.442	0.542	0.687	0.218
Classical belief update	0.513	0.689	0.699	0.838
Recurrent world model	0.000	0.009	0.000	0.254
Persistent memory	0.000	0.009	0.000	0.000
Relational graph	0.252	0.261	0.252	0.204
Structured dynamics	0.259	0.242	0.280	0.218

runs the test suite. The completed full run was executed on a single NVIDIA L4 GPU with 23GB memory and CUDA enabled, using one Python process and cached learned-baseline checkpoints. It took about 11.5 wall-clock hours from the first family evaluation log entry to release-manifest completion. Preliminary/debug runs are not included in the main reported compute unless otherwise stated. The learned training jobs are small; wall time is dominated by task generation, planning/evaluation loops, bootstrap aggregation, and artifact rendering.

C Limitations and future work

MapShift is not a deployment-realism benchmark and should not be interpreted as evidence of broad real-world robustness, language grounding, commonsense reasoning, or deployment readiness. The substrate simplifies perception, semantics, dynamics, and embodiment; severities are ordinal within family; and held-out motifs support leakage-controlled protocol comparison rather than broad motif-generalization claims. Future work should carry CEP and its health checks into richer 3D environments.

D Artifact contents

The release artifact contains versioned configs, split manifests, intervention and task recipes, benchmark health reports, raw per-episode records, family-wise tables, severity-response data, protocol-comparison outputs, rendered paper tables/figures, and top-level manifests. The artifact is an executable benchmark and generator, not a static dataset contribution. Generated environments and tasks can be regenerated from code and configs.

E Broader impact and ethics

MapShift is an evaluation artifact for embodied-planning research. It does not contain human-subject data, scraped web data, personally identifiable information, or deployed decision systems. The main risk is misuse of benchmark scores as evidence for real-world autonomy or robotics safety beyond the benchmark assumptions. The paper mitigates this by stating supported and unsupported

Table A.6: Family-wise mechanism diagnostic scores under CEP. Scores are primary family scores aggregating planning, inference, and adaptation, so the oracle planner is a solvability/planning reference rather than a universal upper bound for every mixed family score. The diagnostic separates stale pre-intervention reuse, local heuristic priors, and explicit belief update.

Method	Metric	Topology	Dynamics	Semantic
Oracle post-intervention planner	0.480	0.745	0.771	0.807
Stale-map planner	0.438	0.447	0.771	0.228
Weak heuristic	0.442	0.542	0.687	0.218
Classical belief update	0.513	0.689	0.699	0.838

Table A.7: Deterministic reference baselines. These methods have no trainable parameters and are included to calibrate solvability, protocol sensitivity, and weak-reference behavior.

Baseline	Access and stored state	Evaluation behavior	Training
Oracle post-intervention planner	Privileged access to the evaluation environment. Used as a solvability and upper-reference check, not as a deployable agent.	Deterministic shortest-path, exact inference answers, and oracle adaptation curve when solvable.	0 params; none.
Same-environment reference condition	Protocol-control wrapper used only when execution is performed in the explored environment rather than the intervened environment.	Shares deterministic planning mechanics with the corresponding evaluation condition; interpreted as a protocol comparison, not a method upper bound.	0 params; none.
Weak heuristic	Stores visited cells, visited node ids, visible goal-token bindings, and limited local traversable structure. No global map revision.	Orthogonal visible-path planner over remembered structure; always predicts no topology change; answers semantic reachability with stale remembered token bindings; predicts masked family by task-type rule; capped heuristic adaptation curve.	0 params; none.
Stale-map planner	Stores the serialized pre-intervention environment from reward-free exploration and deliberately does not update it after intervention.	Plans shortest paths and answers semantic queries using the pre-intervention map, then is scored in the evaluation environment. Used as a diagnostic sensitivity floor for stale exploration-derived state.	0 params; none.
Classical belief update	Stores an occupancy map, edge structure, dynamics signature, and semantic-token bindings from exploration.	Uses local post-shift mismatch observations and the task/adaptation budget to update edges, costs, dynamics, and token bindings, then replans. It does not receive the intervention family label or global post-intervention map.	0 params; none.

claims, reporting benchmark health separately from model results, and emphasizing family-wise diagnostics over a single leaderboard score.

F NeurIPS Paper Checklist

1. **Claims.** Yes. The abstract and introduction state that MapShift is an executable post-intervention evaluation protocol and benchmark, with claims limited to controlled post-intervention embodied planning.
2. **Limitations.** Yes. Appendix C states that MapShift is not a deployment-realism benchmark and should not be interpreted as evidence of broad real-world robustness, language grounding, commonsense reasoning, or deployment readiness.

Table A.8: Learned baseline metadata. Parameter counts are representative realized trainable parameters for the release configuration; exact per-run counts, device metadata, checkpoints, and training summaries are written to run manifests.

Baseline	Stored representation and architecture	Supervision and optimization	Size
Monolithic recurrent world model	Builds node features from the exploration trace and local observations. A GRU with hidden size 12 processes node features and stores a recurrent hidden-state summary.	Self-supervised edge, geometry-cost, traversal-cost, and token-location prediction; 6 epochs; Adam lr 0.01; validation fraction 0.2; early-stopping patience 3; 5 seeds.	1266 params.
Persistent-memory world model	Builds the same graph data, but reads each node through 16 learned memory slots. Slot attention stores persistent exploration-derived state.	Linear node encoder, learned memory slots, attention over slots, and the same self-supervised heads; 8 epochs; Adam lr 0.01; validation fraction 0.2; patience 3; 5 seeds.	358 params.
Relational graph world model	Uses node features and the explored adjacency matrix; applies a node encoder plus 2 message-passing steps over normalized adjacency.	Same self-supervised targets; 10 epochs; Adam lr 0.01; validation fraction 0.2; patience 3; 5 seeds.	396 params.
Structured-dynamics world model	Factorized geometry and dynamics encoders; edge and geometry heads use geometry factors, while traversal-cost prediction additionally receives global dynamics factors.	Same self-supervised targets; 10 epochs; Adam lr 0.01; validation fraction 0.2; patience 3; 5 seeds. Structured by restricting which latent factors feed each prediction head.	504 params.

3. **Theory, assumptions, and proofs.** N/A. The paper formalizes an evaluation protocol and estimands but does not present theoretical theorems requiring proof.
4. **Experimental result reproducibility.** Yes. Methodology, artifact, and appendix sections describe versioned configs, task generation, baseline training, grouped bootstrap estimation, and release artifacts.
5. **Open access to data and code.** Yes. The submission includes an executable artifact containing the benchmark code, versioned configs, reproduction commands, generated outputs, raw per-episode records, health reports, and rendered result tables needed to reproduce the main study. The artifact includes both a smoke-test path and full-study reproduction instructions. The contribution is an executable artifact rather than a static hosted dataset.
6. **Experimental setting/details.** Yes. The paper specifies splits, exploration budget, intervention families and severities, task classes, baseline mechanisms, hyperparameters, model seeds, and aggregation.
7. **Experiment statistical significance.** Yes. The paper specifies 95% bootstrap confidence intervals with 1000 resamples grouped by environment/model-seed units.
8. **Experiments compute resources.** Yes. Appendix B reports the completed full-run setting: a single NVIDIA L4 GPU with CUDA enabled and approximately 11.5 wall-clock hours. It also reports the CPU smoke-test setting and clarifies that preliminary/debug runs are not included in the main reported compute.
9. **Code of ethics.** Yes. The work is designed as a controlled evaluation artifact and does not involve human-subject data, private data, or high-risk deployed systems.
10. **Broader impacts.** Yes. The appendix discusses the main risk: over-interpreting benchmark performance as real-world autonomy readiness.
11. **Safeguards.** N/A. The paper does not release a high-risk pretrained model or dual-use generative model.
12. **Licenses.** Yes. The released code is licensed under the MIT License; generated benchmark outputs and rendered result artifacts are licensed under CC BY 4.0; third-party dependencies and referenced tooling are listed in `THIRD_PARTY_LICENSES.md`.

13. **Assets.** Yes. The paper documents the released benchmark artifact, its generated outputs, limitations, and intended use. No static dataset is introduced.
14. **Crowdsourcing and human subjects.** N/A. The work uses no crowdsourcing and no human subjects.
15. **IRB approvals.** N/A. No human-subject research is conducted.
16. **Declaration of LLM usage.** N/A. LLMs are not a component of the benchmark methodology, baselines, or experimental results.